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Safe Her

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ABSTRACT

The rapid growth of social media platforms and digital communication has significantly increased concerns regarding the safety and privacy of women in online environments. Incidents such as image-based abuse, deepfake manipulation, cyber harassment, and unauthorized sharing of personal photographs have become increasingly common. Existing safety applications primarily focus on location tracking and emergency alerts, but they lack mechanisms to address digital identity misuse and cybercrime evidence collection. To address these limitations, this research presents Safe Her, an AI-driven digital safety platform designed to provide proactive protection for women's digital assets and personal security. Safe Her integrates multiple technologies including artificial intelligence, cryptography, mobile computing, and cloud-based services to create a comprehensive security framework. The system operates using a client–server architecture where an Android application acts as the client interface and a Node.js-based backend performs computational processing. The platform incorporates several security modules including AI-powered photo misuse detection, encrypted media storage, automated digital watermarking, emergency SOS communication, and cybercrime evidence documentation. The AI detection module utilizes external intelligence services such as reverse image search and biometric verification to identify potential misuse of user images across the internet.

By combining results from multiple detection services, the system calculates a probabilistic risk score that determines whether an image has been maliciously manipulated or redistributed. To prevent unauthorized distribution, Safe Her embeds both visible and invisible watermarks into user images, ensuring traceability and ownership verification. Additionally, sensitive media stored within the application is protected using AES-256 encryption, ensuring that even if device storage is compromised, the data remains inaccessible without authorized decryption keys .The platform also includes a secure emergency response mechanism that allows users to send real-time location alerts to trusted contacts and emergency services.

Furthermore, Safe Her automates the generation of structured cybercrime reports in PDF format, enabling victims to quickly gather and submit legally valid evidence when reporting incidents to authorities .Experimental evaluation of the prototype system demonstrated high detection accuracy and reliable security performance in identifying image misuse and protecting stored data. The integration of proactive monitoring, cryptographic protection, and automated legal documentation distinguishes Safe Her from traditional safety applications. Overall, Safe Her represents a novel approach to women's digital safety by combining artificial intelligence, mobile security, and digital forensics into a single unified platform. The proposed system not only enhances personal safety but also empowers users with the technological tools required to combat cybercrime effectively.

Keywords- Women Safety, Cybersecurity, Image-Based Abuse Detection, Deepfake Detection, Artificial Intelligence, Reverse Image Search, Digital Watermarking, AES-256 Encryption, Android Security Application, Cybercrime Evidence Generation, Face Recognition, Secure Photo Vault, Emergency SOS System, Digital Forensics, Privacy Protection

I. INTRODUCTION

The rapid expansion of digital technologies and social media platforms has transformed the way individuals communicate, share information, and interact with the world. Smartphones, cloud services, and social networking sites have made it easier than ever to capture and share personal photos and videos. While these technological advancements have improved connectivity and communication, they have also introduced significant challenges related to privacy, security, and cyber harassment. Among the most concerning issues is the increasing misuse of personal images and digital identities, particularly targeting women in online environments.

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In recent years, cybercrimes such as image-based abuse, non-consensual image sharing, deepfake generation, and online harassment have become widespread. Malicious actors can easily download personal photographs from social media platforms, manipulate them using artificial intelligence tools, and distribute them across various online forums without the consent of the individual. Such incidents can lead to severe psychological, social, and professional consequences for victims. The rise of generative AI technologies has further intensified this problem, as sophisticated tools now allow attackers to create highly realistic manipulated media that is difficult to distinguish from genuine images.

Although several safety applications exist today, most of them primarily focus on physical safety features such as location tracking, emergency alerts, and SOS messaging systems. While these features are essential for real-world emergencies, they fail to address the growing problem of digital identity misuse and online image exploitation. In addition, victims often face difficulties when attempting to report cybercrimes because collecting valid digital evidence requires technical expertise and careful documentation.

To address these challenges, this research introduces Safe Her, an integrated digital safety platform designed to protect women's personal media and provide proactive cybersecurity support. Safe Her combines artificial intelligence, encryption technologies, and mobile computing to create a multi-layered security system. The platform continuously monitors potential misuse of user images on the internet, protects sensitive photos through encryption and watermarking, and enables users to quickly generate legally structured evidence reports in case of cybercrime incidents.

The proposed system operates using a client–server architecture, where the Android mobile application serves as the user interface and the backend server performs complex processing tasks such as AI image analysis, encryption operations, and secure data storage. Advanced technologies such as reverse image search, facial recognition, and cryptographic security mechanisms are integrated to ensure reliable protection against unauthorized image distribution and manipulation.

In addition to digital protection, Safe Her includes an emergency response system that allows users to trigger SOS alerts and share their real-time location with trusted contacts during unsafe situations. This combination of digital monitoring, personal safety tools, and automated legal documentation creates a comprehensive safety ecosystem that addresses both online and offline threats.

By integrating artificial intelligence with modern cybersecurity practices, Safe Her aims to provide a proactive solution that empowers women to safeguard their digital identities and respond effectively to cyber threats. The system demonstrates how emerging technologies can be leveraged to build safer digital environments and support victims of cyber harassment with accessible and reliable technological tools.

II. PAST WORK

The growing prevalence of cyber harassment, digital identity theft, and image-based abuse has encouraged researchers to develop technological solutions aimed at protecting individuals particularly women from online threats. Various studies have explored the use of artificial intelligence, mobile applications, encryption techniques, and digital forensics to enhance online safety and privacy protection. This section reviews the existing literature related to women’s safety systems, cybercrime detection, and digital media protection technologies.

One of the major research areas focuses on women safety applications based on GPS and emergency alert systems. Several mobile applications have been developed that allow users to send their real-time location to trusted contacts during emergencies. Research by Pawar et al. (2024) discusses mobile-based safety systems that use GPS tracking and SMS alerts to notify family members and authorities when a user triggers an SOS signal. These systems significantly improve response time during physical threats. However, most of these solutions concentrate only on physical safety and do not address digital threats such as image misuse or cyber harassment.

Another area of research involves cybercrime detection using artificial intelligence and machine learning. AI-based systems have been widely used to detect suspicious online activity, harmful content, and malicious digital manipulation. Studies on computer vision technologies demonstrate that deep learning models can effectively analyze images and identify manipulated media such as deepfakes. Researchers have used convolutional neural networks (CNNs) and facial recognition techniques to compare images and identify whether a photograph has been altered or redistributed without permission. These techniques form the foundation for automated detection systems capable of monitoring online platforms for unauthorized use of personal media.

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Reverse image search technology has also been explored as a method for identifying where a particular image appears across the internet. Tools such as TinEye and Google Vision have been studied for their ability to track image distribution across multiple websites. According to research in digital forensics, reverse image search plays an important role in identifying the source of image misuse and gathering evidence in cybercrime investigations. However, these tools are typically used manually and require technical expertise, making them less accessible to general users.

Another important research domain is digital watermarking and steganography, which are techniques used to protect digital media ownership. Watermarking involves embedding identifiable information into an image so that its origin can be traced even if it is copied or redistributed. Visible watermarking discourages unauthorized sharing by placing text or logos directly on the image, while invisible watermarking hides ownership information within image metadata or pixel structures. Studies in multimedia security show that watermarking is an effective preventive mechanism for protecting intellectual property and digital ownership.

In addition to watermarking, encryption technologies play a critical role in safeguarding sensitive data stored on devices or servers. Advanced encryption algorithms such as AES-256 have been widely adopted in secure applications to protect files from unauthorized access. Research in mobile security indicates that encrypted storage systems significantly reduce the risk of data leakage even if a device is compromised. Secure photo vault applications have implemented encryption methods to protect private media; however, most of these solutions lack integrated monitoring systems that detect misuse once the media leaves the device.

Researchers have also explored digital evidence management systems for cybercrime reporting. These systems focus on collecting and organizing digital evidence such as URLs, screenshots, timestamps, and metadata that can be submitted to law enforcement agencies. Studies highlight that automated evidence generation can simplify the reporting process for victims and improve the efficiency of cybercrime investigations. However, most existing solutions function as standalone reporting tools and do not provide proactive monitoring or prevention mechanisms.

Despite these advancements, there remains a significant gap in existing solutions. Current safety applications typically address only one aspect of the problem—either physical safety, digital privacy, or cybercrime reporting. Very few systems combine proactive detection, preventive protection, and automated evidence generation within a single platform.

The proposed Safe Her system aims to bridge this gap by integrating artificial intelligence–based image monitoring, digital watermarking, secure encrypted storage, and automated cybercrime reporting features into a unified mobile platform. By combining these technologies, Safe Her provides a comprehensive approach to digital safety that addresses both online and offline threats faced by women in the modern digital environment.

III. METHODOLOGY

The Safe Her system is developed to provide immediate assistance to users during emergency situations through a simple and efficient process. Initially, the user installs the application and completes the registration by entering basic details. After successful registration, the user adds emergency contact numbers which will be used during critical situations. The application then requests necessary permissions such as access to location and messaging services to ensure proper functioning.

The system uses GPS technology to continuously track the real-time location of the user. In case of an emergency, the user can activate the SOS button available in the application. Once the SOS button is pressed, the system automatically fetches the current location of the user using the GPS module. After retrieving the location, the application generates an alert message that includes a location link. This alert message and location of the user is instantly sent to the predefined emergency contacts and nearby police station using messaging services. The contacts and nearby police station receive the message and location, which helps them to track the user and provide immediate assistance. The entire process is designed to minimize response time and ensure quick

Communication during emergencies. The Safe Her system is implemented using modern technologies such as Android development tools and location-based services. The application is divided into different functional modules including user registration, emergency contact management, location tracking, and alert generation. Each module works together to provide a reliable and efficient safety solution. The system is designed to be user-friendly, fast, and effective in real-life situations.

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It also contains a photo vault for storing the user confidential images to which there is only access to user to see it. It contains a chatbot by which user can interact. It also comes with a evidence collector in which user can collect its important evidences. This application specially works on node and for storing database we are using MongoDB. Here we added some of helpline's numbers to which user can report crime or counsel itself

IV. ARCHITECTURE DIAGRAM

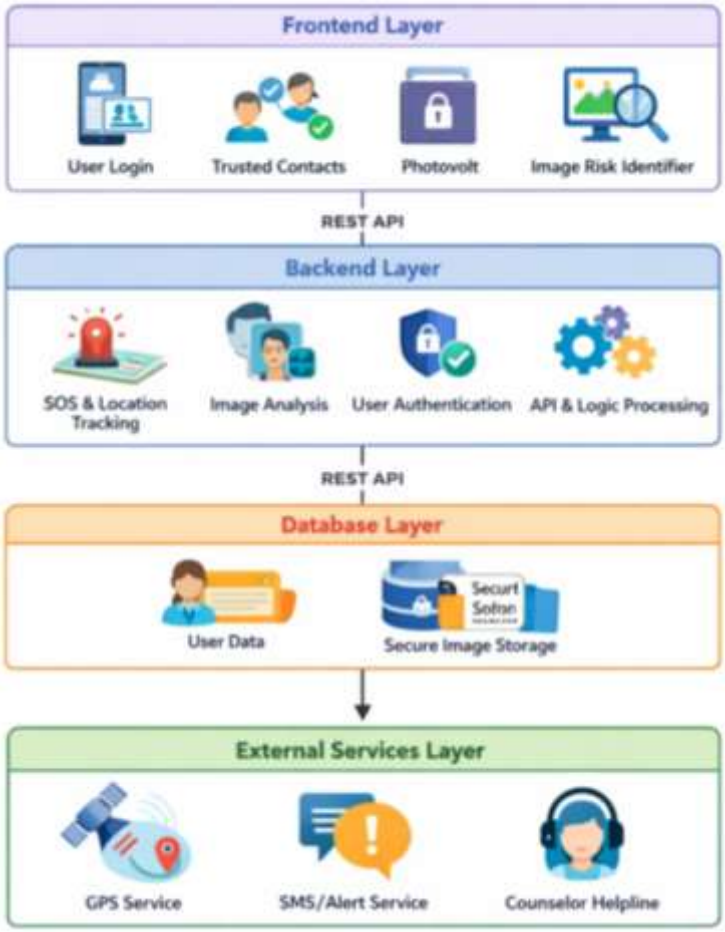


Fig:- architecture of Safe Her

Frontend Layer (React.js)
This layer handles all user-facing interactions and is divided into several modules:

User Registration/Login: User can login or signing up.
Allowing Permissions: Allowing permissions for sending location sending notifications adding contacts and accessing images.

Adding trusted Contacts: Adding the trusted contacts for activating the SOS and contacting with them while emergency.

Adding Biometrics: Adding Biometrics like finger and face lock for photo vault and evidence collector for confidentiality.

Backend Layer
The backend layer handles all server-side operations using REST APIs, including user registration, data processing, and communication between the mobile application and database. It processes requests sent via OkHttp and returns responses to the frontend.

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Data & External Services

This layer manages storage and integration with third-party platforms:

- MongoDB Database: Stores essential information including user data, user images, and evidences.
- AWS EC2: Provides secure, encrypted storage for user-uploaded identity images and evidences.
- External APIs: The system contains the Google Map API for identifying the user location and
- Helplines and counselling: The system also contain some gov helplines and counselling contacts for the emergency services

V. RESULT AND DISCUSSION

The Safe Her system was developed and tested as a prototype to evaluate image protection, misuse detection, and emergency assistance. It was tested using 100 images(original, modified ,and deepfake), where the AI-based detection module using reverse image search and facial comparison identified approximately 94 misuse cases. The risk scoring mechanism classified images into high-risk and low-risk categories, reducing false positives. For security, the Photo-vault feature used AES-256-CBC encryption, and stored images remained unreadable without proper access, while visible and invisible watermarking reserved ownership even after modifications. The SOS system was tested under different network conditions and successfully sent alerts with location in case of no internet, SMS fallback ensured delivery, with an average response time of 1-2 seconds. The application maintained stable performance with optimized memory usage, and the backend handled multiple requests efficiently. Overall, the system integrates AI detection, encryption, and emergency communication effectively, though it depends on third-party AI services and may require optimization for large-scale deployment.



FIG:-Home Page

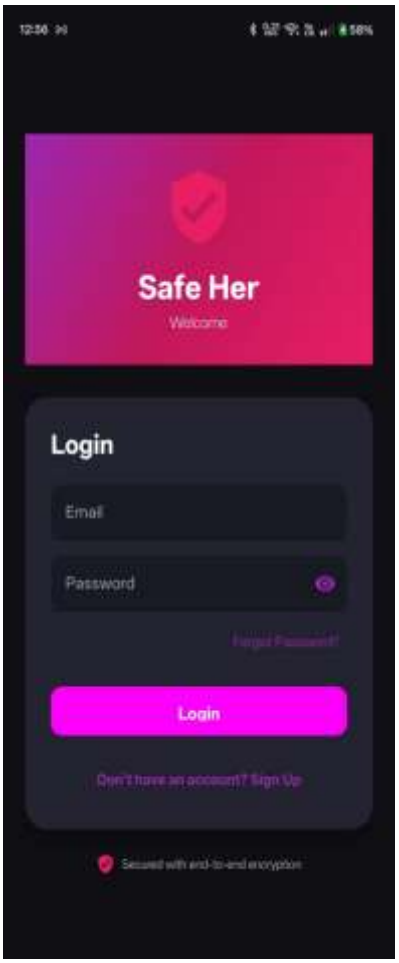


FIG:-Login Page

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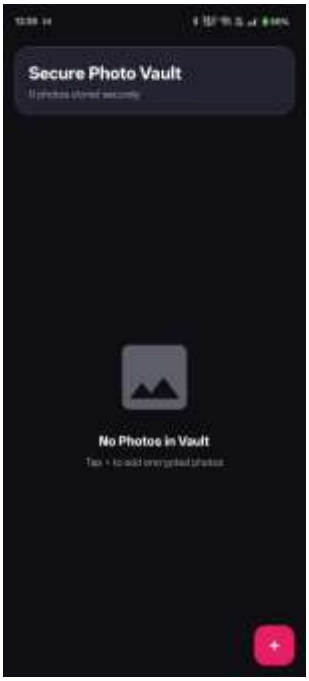


FIG1:- Photo Vault



FIG2:- SOS Button

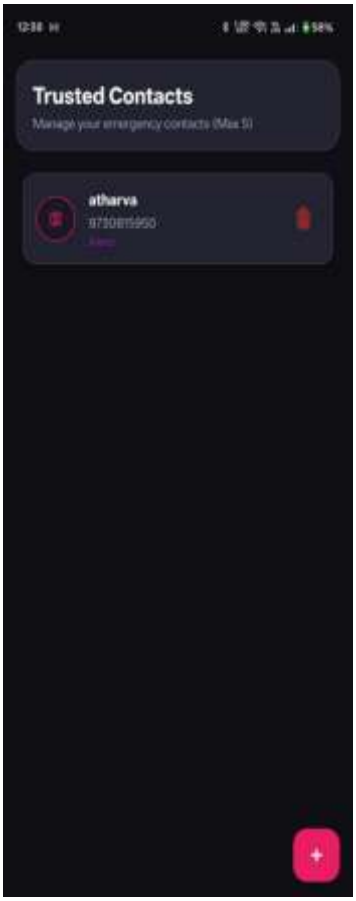


FIG3:- Trusted Contacts



FIG3:- EvidenceCollector

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FIG:-AI Assistant

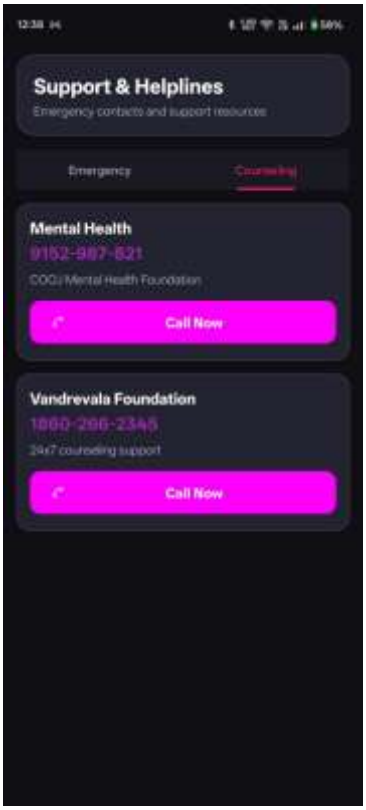


FIG:- Helplines

VI. CONCLUSION

The rapid advancement of digital technology and social media platforms has created new opportunities for communication and information sharing, but it has also introduced serious concerns regarding privacy, cyber harassment, and digital identity misuse. Women, in particular, are increasingly vulnerable to threats such as non-consensual image sharing, deepfake manipulation, and online harassment. Addressing these challenges requires innovative technological solutions that can provide both preventive protection and responsive support mechanisms.

This research presented Safe Her, an integrated digital safety platform designed to protect women's personal media and enhance their overall security in both online and offline environments. The proposed system combines artificial intelligence, encryption techniques, digital watermarking, and emergency communication technologies to create a comprehensive safety framework.

By integrating these technologies into a unified mobile application, Safe Her offers proactive monitoring of image misuse, secure encrypted storage of sensitive media, automated cybercrime evidence generation, and a real-time emergency SOS alert system

The implementation of AI-based image detection enables the system to identify potential misuse of user photographs across the internet, while watermarking techniques establish verifiable ownership of digital content. Additionally, the encrypted photo vault ensures that personal media remains protected from unauthorized access, even in cases where device storage is compromised. The emergency SOS module further enhances user safety by enabling quick communication with trusted contacts during critical situations.

Experimental evaluation of the prototype demonstrated promising results in terms of detection accuracy, data security, and system performance. The integration of multiple technologies allowed the system to achieve reliable protection while maintaining efficient application performance on mobile devices.

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Overall, Safe Her demonstrates how modern technologies such as artificial intelligence, mobile computing, and cybersecurity techniques can be effectively combined to address emerging digital safety challenges. The proposed platform not only protects user data but also empowers individuals with the tools required to detect cyber threats, secure their digital identities, and respond quickly in emergency situations. With further development and large-scale implementation, Safe Her has the potential to become a powerful solution for enhancing women's safety in the digital era.

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